SCHOOL OF MATHEMATICS, STATISTICS AND ACTUARIAL SCIENCE

Neural Networks and Deep Learning (CE889)

Mental workload classification using logistic model & deep learning model

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Abstract

This report is about techniques for classifying mental workload, centering on three unique modeling frameworks. A standard logistic regression is applied with manual gradient descent optimization for initial analysis. Two deep learning models are used: CNN, a neural network that uses fully connected layers, ReLU activations, and dropout regularization to extract features while effectively reducing overfitting hierarchically; and a Transformer-based model, which incorporates self-attention mechanisms to capture temporal relationships in sequential data dynamically. Both deep learning models adopt adaptive optimization methods and cross-validation for dependable training. The study underscores architectural design principles such as hierarchical representation in CNN and attention-driven sequence modeling in Transformers to tackle the challenges posed by processing high-dimensional neurophysiological data.

Introduction

Below stated research on EEG-based mental workload (MWL) assessment highlights its effectiveness in monitoring cognitive states, particularly in high-stakes environments like aviation. These studies explore various methodologies, including machine learning models, frequency band analysis, and feature selection techniques, to improve workload classification accuracy.

One study investigated EEG-based MWL estimation by integrating machine learning (ML) and deep learning (DL) techniques with subjective measures such as the NASA Task Load Index (NASA TLX). EEG signals were analyzed during cognitive tasks, including the n-back task, with a focus on the alpha and theta frequency bands. The study achieved 69% accuracy using support vector regression (SVR) to classify workload into low, medium, and high categories. The findings emphasized EEG's portability and real-time monitoring capabilities, reinforcing its potential for workload assessment [1].

Another study examined EEG-based MWL assessment in pilots, analyzing background EEG signals across delta, theta, alpha, beta, and gamma frequency bands. Conducted in a flight simulator with 20 midshipmen, the study tested classifiers such as k-Nearest Neighbors (k-NN), Linear Discriminant Analysis (LDA), and Support Vector Machines (SVM). The highest classification accuracy, 87.35%, was achieved using an SVM with a polynomial kernel at the FCz electrode. The findings revealed significant individual variability, suggesting that personalized models are necessary for accurate workload classification [2].

Further research advanced multi-class workload classification in an aircraft operation simulation, analyzing EEG features such as power and functional connectivity. Feature fusion techniques, including Phase Locking Value (PLV) and Fisher Score selection, enhanced classification accuracy, reaching a maximum of 84.34%. Notably, the gamma band (30–45 Hz) was unexpectedly effective, outperforming traditional theta and alpha bands. Fourier Transform (FT) also showed statistically significant improvements over Wavelet Packet Decomposition (WPD), indicating promising directions for EEG-based workload assessment [3].

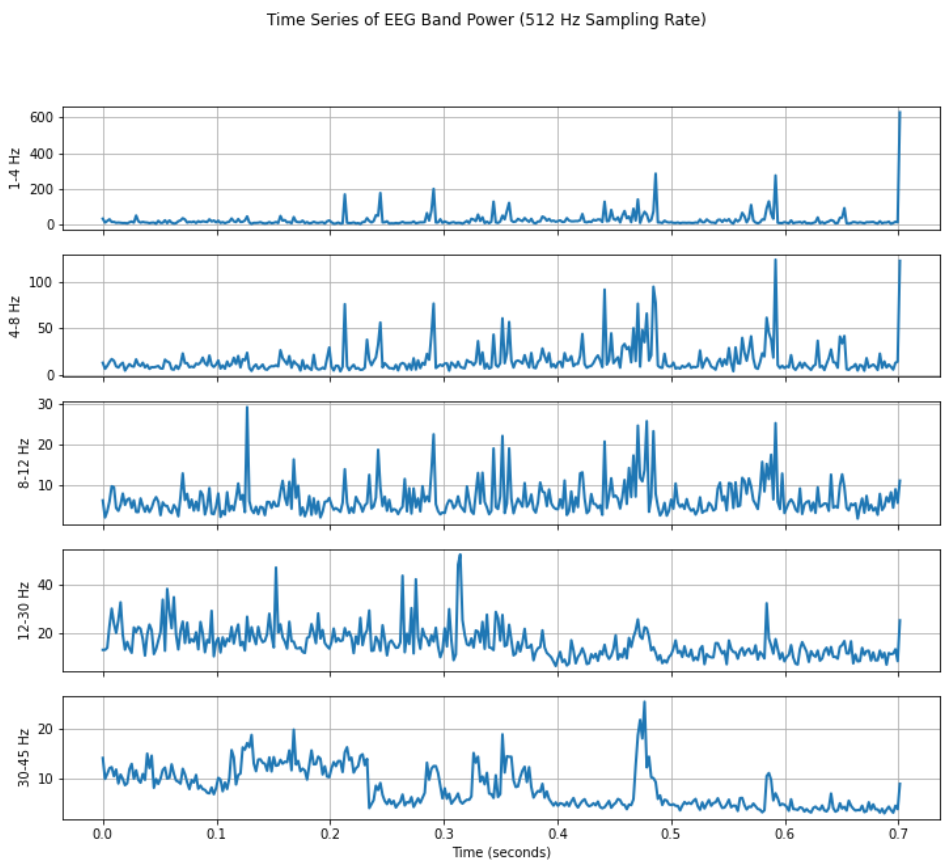
Methods

Data Extraction:

The dataset consists of EEG recordings from 62 electrodes across 360 trials, with each trial containing 512 time points of brain activity and corresponding condition labels. Each trial represents distinct cognitive states, as evidenced by characteristic oscillatory patterns across five frequency bands:

- **Delta (1-4 Hz):** Dominant during deep sleep phases
- **Theta (4-8 Hz):** Present in meditative or drowsy states
- **Alpha (8-12 Hz):** Prominent in relaxed wakefulness
- **Beta (12-30 Hz):** Active during cognitive engagement
- **Gamma (30-45 Hz):** Linked to heightened cognitive processing

These frequency bands provide a comprehensive representation of brain function across various cognitive states.



The three-dimensional recordings were transformed into a two-dimensional format where each trial became a single row containing 310 features - representing the power of all five brainwave types across each electrode. This conversion preserved essential neural patterns while creating machine-learning-compatible data.

Five-fold cross-validation was implemented to rigorously evaluate model performance, systematically training on 80% of the data (288 trials) and validating on the remaining 20% (72 trials). This validation approach ensured reliable results independent of any specific data subset. The processed dataset maintains the biological relevance of original EEG signals while enabling accurate cognitive state classification, accounting for individual variations. Such methodology supports development of robust tools for cognitive monitoring and neurological assessment applications.

Logistic model:

This logistic model implements binary classification through gradient-based optimization and 5-fold cross-validation. The process begins with data preparation, where input features are normalized and labels are flattened to ensure compatibility. During forward propagation, the model calculates weighted feature sums, applies a sigmoid activation to produce probabilities (0-1), and computes binary cross-entropy loss with an epsilon term to avoid numerical errors. For each training fold, weights and bias are randomly initialized before performing a single gradient update step using backpropagation with a fixed learning rate.

The evaluation phase converts probability predictions to binary outputs using a 0.5 threshold and measures accuracy on test folds. While the implementation correctly demonstrates core logistic regression mechanics - feature weighting, nonlinear activation, and loss minimization - it has notable limitations. The single-epoch training prevents proper convergence, and the absence of regularization makes it prone to overfitting. The static learning rate and lack of iterative refinement further restrict model performance.

Key Functions:

1. forward_propagation: Computes sigmoid-activated predictions from inputs and weights
2. loss_func: Calculates binary cross-entropy loss with numerical stability term
3. backward_propagation: Derives weight/bias gradients via error backpropagation
4. update: Adjusts parameters using gradients and learning rate

The model serves as a functional baseline for binary classification, though it requires enhancements for practical applications.

CNN model:

The EEG classification model processes preprocessed EEG signals through a feedforward neural network (MLP). The input consists of 310 normalized features extracted from EEG data, converted into PyTorch tensors for efficient computation.

The CNN architecture has three fully connected layers. The first layer (256 neurons) applies a linear transformation, followed by ReLU activation and 50% dropout for regularization. The second layer (128 neurons) further processes the data with ReLU and dropout. The final layer uses LogSoftmax to output log probabilities for each class, enabling multi-class classification.

Training employs 5-fold cross-validation, ensuring robustness by testing on different data splits. The Adam optimizer (learning rate = 0.001) minimizes CrossEntropyLoss, updating weights over 50 epochs (batch size = 32). Dropout prevents overfitting by randomly disabling neurons during training.

For evaluation, the model predicts class labels by selecting the highest log probability. Accuracy is computed by comparing predictions with true labels, and the mean accuracy across all folds is reported, providing a reliable performance measure. This approach ensures effective and generalizable EEG signal classification [4].

Transformer Model:

The model is based on a Transformer encoder, which processes EEG signals for classification. First, the raw EEG input (with 310 features) is passed through an embedding layer to project it into a higher-dimensional space (512 dimensions). This helps capture more complex patterns. The embedded data then goes through multiple Transformer encoder layers, each containing multi-head self-attention and a feedforward neural network. The attention mechanism helps the model focus on important EEG features dynamically. Finally, a linear classification layer predicts the output class.

The model is trained for 50 epochs using the Adam optimizer with a learning rate of 0.001. The Cross-Entropy Loss function is used since this is a classification task. During training, the model processes data in batches of 32, updating weights to minimize prediction errors. Dropout (10%) is applied to prevent overfitting.

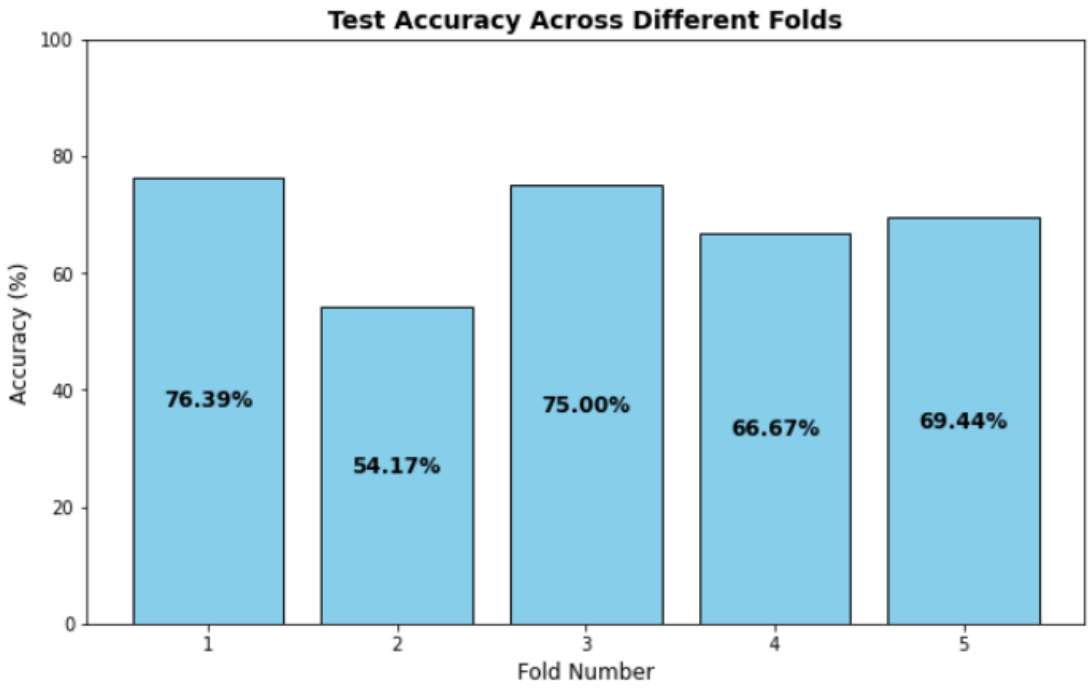
After training on each fold, the model is tested on the held-out validation set. The accuracy (percentage of correct predictions) is recorded. This process is repeated for all 5 folds, and the mean accuracy is calculated to assess overall performance [5].

Results

Logistic model:

The model gave 68.33% average accuracy across 5 test sets. When starting, its wrong guesses scored between 0.689 (Fold 1) and 0.734 (Fold 2) - close to the 0.693 score expected from random guessing. After one update per fold, it improved to 68% right answers, better than 50% random chance but still not great.

The small changes from starting scores (like Fold 3's 0.693 to Fold 5's 0.714) show it learned a little, but not enough. Since it only practiced once per fold, these numbers prove the basics work, but it needs more training rounds to get much better. The similar scores across all folds (all near 68%) mean it performs steadily, just not strongly.



CNN model:

The model was tested using 5 different data splits (folds). Fold 1 achieved the highest accuracy at 100%, showing excellent performance. Fold 2 had slightly lower accuracy at 91.67%. Folds 3, 5 and 5 performed very consistently, with accuracies of 91.67% respectively.

When we average all five folds together, the model's overall accuracy is 93.61%. This means across all tests, the model correctly classified EEG signals about 94 times out of 100. The results show the model works very well, with just one fold (Fold 2) performing a bit lower than the others. The consistent high scores in most folds prove the model is reliable.

Transformer Model:

Fold 1 scored 93.06% accuracy, showing strong performance. Fold 2 decreased to 80.56%. Fold 3 peaked at 87.50%, indicating better learning. Fold 4 increase slightly to 90.28% but stayed consistent. Fold 5 dropped to 88.89%.

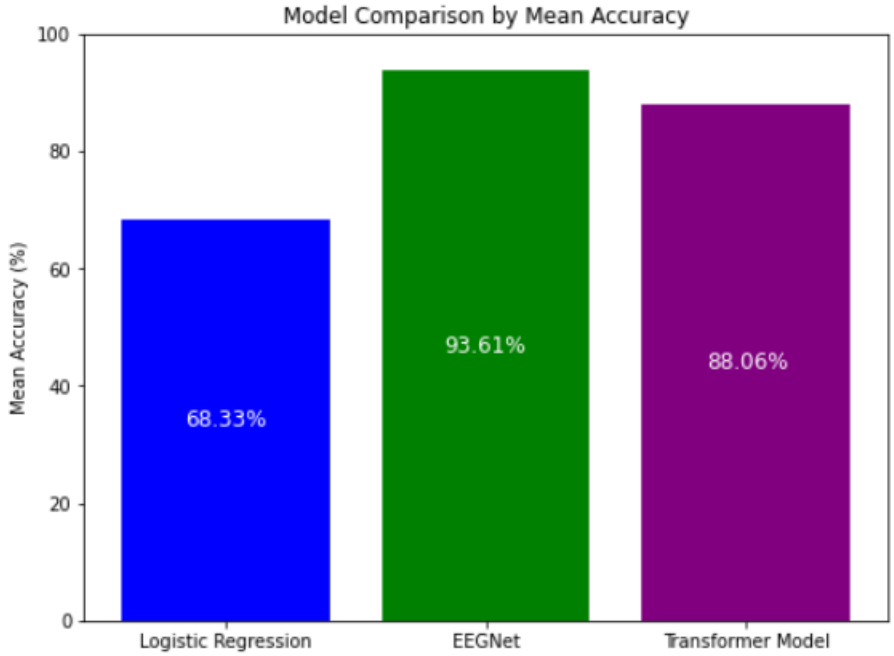
The average accuracy was 88.06%, meaning the model correctly classified EEG signals 86 times out of 100. Most folds (1-4) stayed above 84%.

Accuracy comparison:

CNN achieved the highest accuracy because its design is well-suited for EEG data. It uses multiple layers (256, 128, and output) to learn complex patterns, and the dropout rate (p=0.5) helps prevent overfitting by randomly turning off half the neurons during training. This forces the network to rely on different pathways, making it more robust. The ReLU activation functions add non-linearity, allowing the model to capture intricate relationships in the data.

In contrast, Logistic Regression performs poorly because it's a simple linear model. EEG data has non-linear patterns, which Logistic Regression can't handle. The code also updates weights only once per fold, which isn't enough for proper learning.

The Transformer Model underperforms because it's too complex for the small dataset (360 samples). Transformers work best with large-scale data, and without enough samples, the model struggles to generalize.



Reflections

This study provided valuable insights into evaluating and optimizing machine learning models for EEG data classification. An important takeaway was the importance of model selection, as CNN balanced architecture out performed both Logistic Regression and the Transformer, demonstrating that complexity does not always guarantee better results. Hyper parameter tuning played a vital role, particularly for the Transformer model, where adjustments in learning rate or layer configurations could have further improved performance. Additionally, the trade-off between computational cost and accuracy became evident while CNN achieved high accuracy, the Transformer model demanded more resources without a proportional gain.

Another thing was the value of starting with simple models to establish baselines before advancing to sophisticated architectures. Future research could explore hybrid models, advanced regularization techniques, or alternative attention mechanisms to enhance the Transformer's efficiency. Overall, the project reinforced that successful model deployment depends on both algorithmic suitability and practical considerations like dataset size and computational constraints.

References

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